

Land Use / Land Cover (LULC) Classification

Experiment 4: Multi-Seasonal Features with Increased Sampling

Random Forest vs. XGBoost — 4-Season Bands, Larger Sampling & Spectral Indices

Project Report

1. Objective

The objective of this experiment was to investigate the effect of multi-seasonal spectral information, increased sample size, and spectral-index-based feature augmentation on LULC classification performance.

A large and balanced dataset containing 90,000 samples, with 10,000 samples per class across nine LULC classes, was used. Four seasonal observations were incorporated to capture temporal spectral variation. The experiment was conducted in two feature configurations:

- Raw multi-seasonal spectral bands
- Raw multi-seasonal spectral bands + spectral indices

Random Forest was evaluated on both configurations, while XGBoost was additionally evaluated on the raw multi-seasonal configuration.

2. Dataset Configuration

Parameter	Configuration
Experiment ID	EXP-04
Experiment Type	Multi-Season + Increased Sampling + Feature Augmentation
Total Samples	90,000
Samples per Class	10,000
Number of Classes	9
Seasons	4 (Q1, Q2, Q3, Q4)
Test Samples	18,000
Test Set	20%
Class Labels	10, 20, 30, 40, 50, 60, 80, 90, 95

The dataset was perfectly balanced, with 10,000 samples for each of the nine LULC classes.

3. Feature Configurations

Configuration A — Raw Multi-Seasonal Bands

Ten Sentinel-2 spectral bands were used for each of the four seasons: B2, B3, B4, B5, B6, B7, B8, B8A, B11, B12.

Therefore: 10 bands $\times$ 4 seasons = 40 features. No spectral indices or additional feature engineering were applied.

Configuration B — Raw Bands + Spectral Indices

The same 40 raw multi-seasonal features were augmented with seven spectral indices: NDVI, NDRE, NDBI, BSI, NDWI, MNDWI, and EVI.

Each index was calculated for all four seasons: 7 indices $\times$ 4 seasons = 28 index features.

Therefore: 40 raw bands + 28 spectral indices = 68 features. No temporal difference features were included in this configuration.

Feature Summary

Feature Group	Configuration A	Configuration B
Raw spectral bands	40	40
Spectral indices	—	28
Total features	40	68

4. Class Distribution

Class	Samples
10	10,000
20	10,000
30	10,000
40	10,000
50	10,000
60	10,000
80	10,000
90	10,000
95	10,000
Total	90,000

5. Models Evaluated

The following models were evaluated:

- Random Forest
- XGBoost

The models were trained using the same dataset and class distribution.

6. Overall Results

Feature Configuration	Model	Accuracy	Macro Precision	Macro Recall	Macro F1	Weighted F1
40 Raw Bands	Random Forest	87.00%	0.87	0.87	0.87	0.87
40 Raw Bands	XGBoost	88.21%	0.88	0.88	0.88	0.88
40 Raw Bands + 28 Indices	Random Forest	87.00%	0.87	0.87	0.87	0.87

Feature Configuration	Model	Accuracy	Macro Precision	Macro Recall	Macro F1	Weighted F1
40 Raw Bands + 28 Indices	XGBoost	—	—	—	—	—

Best Result

The best-performing configuration evaluated in Experiment 4 was: 40 multi-seasonal raw bands + XGBoost, with:

- Accuracy = 88.21%
- Macro F1 = 0.88

7. Random Forest — Configuration A

40 Raw Multi-Seasonal Bands

Class	Precision	Recall	F1-score	Support
10	0.82	0.82	0.82	1,999
20	0.84	0.85	0.85	1,997
30	0.79	0.80	0.80	1,967
40	0.90	0.92	0.91	1,970
50	0.85	0.82	0.83	2,060
60	0.88	0.86	0.87	2,038
80	0.98	0.99	0.99	1,971
90	0.88	0.83	0.85	2,125
95	0.90	0.96	0.92	1,873
Macro Avg	0.87	0.87	0.87	18,000
Weighted Avg	0.87	0.87	0.87	18,000

8. XGBoost — Configuration A

40 Raw Multi-Seasonal Bands

Class	Precision	Recall	F1-score	Support
10	0.83	0.82	0.83	2,000
20	0.85	0.86	0.85	2,000
30	0.82	0.81	0.82	2,000
40	0.93	0.91	0.92	2,000
50	0.85	0.87	0.86	2,000

Class	Precision	Recall	F1-score	Support
60	0.87	0.89	0.88	2,000
80	0.99	0.98	0.99	2,000
90	0.84	0.88	0.86	2,000
95	0.96	0.91	0.94	2,000
Macro Avg	0.88	0.88	0.88	18,000
Weighted Avg	0.88	0.88	0.88	18,000

9. Random Forest — Configuration B

40 Raw Bands + 28 Spectral Indices

Class	Precision	Recall	F1-score	Support
10	0.82	0.82	0.82	1,997
20	0.84	0.85	0.84	1,965
30	0.79	0.79	0.79	1,992
40	0.90	0.92	0.91	1,947
50	0.85	0.82	0.84	2,056
60	0.89	0.87	0.88	2,057
80	0.98	1.00	0.99	1,969
90	0.88	0.84	0.86	2,117
95	0.90	0.95	0.93	1,900
Macro Avg	0.87	0.87	0.87	18,000
Weighted Avg	0.87	0.87	0.87	18,000

10. Effect of Adding Spectral Indices

This is one of the most important findings from Experiment 4. For Random Forest:

Configuration	Features	Accuracy	Macro F1
Raw multi-seasonal bands	40	87.00%	0.87
Raw bands + spectral indices	68	87.00%	0.87

Adding 28 spectral-index features did not produce a measurable improvement in overall Random Forest performance. Accuracy remained at 87.00%, and Macro F1 remained at 0.87.

However, some class-level differences were observed. For example, Class 95 improved from 0.92 to 0.93 F1, while Class 30 decreased from 0.80 to 0.79.

11. Effect of Multi-Seasonal Information

A particularly useful comparison is between EXP-03 and EXP-04. Both experiments contain 90,000 samples, 9 classes, raw spectral bands, and Random Forest / XGBoost evaluations. The main difference is the temporal feature representation.

Metric	EXP-03	EXP-04
Seasons	1	4
Raw-band features	10	40
Samples	90,000	90,000
Classes	9	9
RF Accuracy	85.00%	87.00%
XGB Accuracy	84.74%	88.21%
RF Macro F1	0.85	0.87
XGB Macro F1	0.85	0.88

For XGBoost, the increase was from 84.74% to 88.21%, or approximately +3.47 percentage points. This demonstrates that incorporating observations from multiple seasons provided additional discriminative information under this experimental setup.

12. Effect of Increased Sample Size

EXP-01 and EXP-04 also provide a useful comparison because both use the same basic 40-feature, four-season raw-band representation.

Metric	EXP-01	EXP-04
Features	40	40
Seasons	4	4
Classes	9	9
Samples	27,000	90,000
RF Accuracy	85.00%	87.00%
XGB Accuracy	87.02%	88.21%

The increase in dataset size was associated with an improvement in performance for both models under the four-season configuration. However, this should be interpreted as an observed experimental relationship, since the experiments were not designed as a controlled statistical ablation study.

13. Key Findings

1. Multi-seasonal information was beneficial

- Moving from a single-season representation to four seasons improved classification performance, particularly for XGBoost.

2. Larger sampling improved the four-season configuration

- Increasing the dataset from 27,000 to 90,000 samples while retaining the 40-feature multi-season representation improved both RF and XGBoost performance.

3. Spectral indices did not improve Random Forest overall accuracy

- Adding 28 spectral-index features increased dimensionality from 40 to 68, but RF accuracy and Macro F1 remained at 87% and 0.87, respectively.

4. XGBoost was the strongest model in this experiment

- The highest observed performance was XGBoost + 40 multi-seasonal raw bands = 88.21% accuracy and 0.88 Macro F1.

14. Conclusion

Experiment 4 investigated the effect of multi-seasonal spectral information, increased sample size, and spectral-index augmentation on LULC classification. A balanced dataset of 90,000 samples across nine LULC classes was used, with four seasonal observations providing 40 raw spectral features per sample.

Random Forest achieved 87.00% accuracy, while XGBoost achieved the highest performance of 88.21% with a Macro F1-score of 0.88. An additional Random Forest experiment incorporating 28 spectral-index features increased the feature space from 40 to 68 features but did not result in an improvement in overall accuracy or Macro F1-score, both remaining at 87% and 0.87, respectively.

Comparison with the single-season configuration demonstrated the benefit of multi-seasonal information, with XGBoost improving from 84.74% to 88.21%. Overall, the results indicate that a large balanced dataset combined with multi-seasonal raw spectral information provided the strongest configuration evaluated in this study, while the addition of spectral indices did not provide a measurable overall improvement for Random Forest.